Experimental Fixes and Paper Figures

Validation, reliability, and a fix for the false-positive weakness in the UAV federated-learning backdoor defense

Will Jedrzejczak **Dilpreet Gill** **Cole Walther**

Group 1 | James Madison University Capstone (IT 445 / IT 499)

In plain terms: what this is and what we found

We are building a GPS spoofing detector that runs across a fleet of drones. Each drone trains its own copy of a small detector on the GPS signals it sees, and a central server merges those copies every round, so no raw data ever leaves a drone. The danger we study is that two of the ten drones have been taken over. They quietly feed their copy bad training examples so that the merged detector learns a secret blind spot: whenever a particular signal value appears, call the signal safe. Later the attacker can stamp that value on a real spoofing attack and the whole fleet waves it through. Our defense is a referee that sits on the server. Before merging, it quizzes every drone model on data the server itself holds, and quietly reduces the influence of any drone whose answers look wrong in a very specific way.

Last report showed the defense worked, but on a single random run. This report does three things: it fixes the errors that were pointed out, it reruns everything three times to prove the result was not luck, and it fixes the one real weakness those tougher checks uncovered. The short version is that the defense holds up, and it is now much fairer to the innocent drones than it was.

What changed this week

- Corrections: the model parameter count was wrong and is fixed. We also proved, rather than assumed, that the data the referee uses is completely separate from the training and test data, and that check caught a real (tiny) data leak, which we fixed.
- Reliability: every main experiment was rerun on three random seeds and is reported as an average with a standard deviation, instead of a single run.
- A real improvement: the tougher checks showed the referee was wrongly suspecting innocent drones 20.5 percent of the time. We fixed that. It is now 0.3 percent, and every other number got better at the same time.

The recommended setting

Use configuration D2: a gentler penalty (beta = 1.0) plus a "dead-zone" so small amounts of suspicion are ignored entirely (tau = 2.0), on top of the existing median-based merging. It cuts wrongly-suspected innocent drones from 20.5 percent to 0.3 percent, still catches the attackers 100 percent of the time, still neutralizes the backdoor, and raises both clean accuracy and spoofing recall. It is a two-line change. Note that Table 1 is still reported at the older beta = 2.0 so it stays directly comparable with the last report; adopting D2 in the headline table is the next small task.

Two results here are less flattering than last time, on purpose. Running three seeds showed that "zero false positives" and "the defense costs nothing" were both too optimistic. We corrected them and then fixed the underlying problem rather than leaving it for a reviewer to find.

1. Correction note

Model parameter count: 3,329 is correct

The notebook now counts the parameters live, layer by layer, instead of quoting a number. The model has 3,329 parameters, which is 13.0 KB stored as float32. The earlier "about 13,000 parameters" was copied from the Week 4 WSN-DS model, which genuinely has 12,805, and the 13.0 kilobyte size made the wrong figure look plausible. Kilobytes are not parameters. Corrected everywhere it appeared.

Root/challenge set separation: proven, and a real leak was found and fixed

We hash every row and count how many appear in more than one partition, rather than asserting separation. The server referee set was already clean. The check did find a 1-row overlap between the client training pool and the test set. The cause: de-duplication ran before the PRN, RX, and TOW identifier columns were dropped, so two rows that differed only in satellite, receiver, or timestamp became identical once those columns were removed, and survived. Fix: de-duplicate on the 10 model input features after dropping identifiers. That removes 2 rows out of 470,546 and makes all three partitions provably disjoint. The notebook now asserts this and stops if it is ever violated. All results were recomputed after the fix.

Pair	Shared rows
server referee set vs client training pool	0
server referee set vs test set	0
client training pool vs test set (after fix)	0

Table A. Row-hash overlap between the three data partitions. All zero after the fix.

Reproducibility

Every table and figure in this report is produced by a notebook cell and exported as a CSV or PNG. The PDF is assembled by a script that reads those exported files and embeds the saved figures, so the report cannot disagree with the experiment. Nothing is transcribed by hand.

Overstated claims, corrected

- "Zero false positives" is retracted. Across three seeds the rate is 20.5 percent and one honest client-round reached zero trust. Section 3 then reduces the real rate to 0.3 percent.
- The negative backdoor lift reported last time was single-seed noise. The three-seed estimate is slightly positive, +0.0107 +/- 0.0025.
- "No utility cost" is softened: clean accuracy under defense is about 0.008 below the honest baseline, small but consistent.
- "Feature-agnostic" is narrowed to "agnostic within the discriminative feature set." A trigger hidden in a feature with almost no signal would fall outside what the referee probes.

2. Reliability: three seeds, and the improved main table

Table 1. Main comparison across three random seeds (mean +/- standard deviation)

Case	Clean Accuracy	Spoofing Recall	BSR	Backdoor Lift
Honest FedAvg	0.7112 +/- 0.0019	0.5292 +/- 0.0115	0.6367 +/- 0.0141	0.0000 +/- 0.0000
Attack (FedAvg)	0.6932 +/- 0.0032	0.3641 +/- 0.0096	0.8782 +/- 0.0178	0.2415 +/- 0.0048
Attack + inflation (Acc-Weighted)	0.6897 +/- 0.0046	0.3524 +/- 0.0148	0.9402 +/- 0.0096	0.3036 +/- 0.0124
Full defense	0.7029 +/- 0.0031	0.5204 +/- 0.0020	0.6474 +/- 0.0156	0.0107 +/- 0.0025

Table 1. Honest baseline, attack, attack with accuracy inflation, and the full defense, over seeds 42, 7, and 123. Columns: clean accuracy on the untouched test set; spoofing recall with no trigger applied; backdoor success rate (BSR) on the triggered test set; and backdoor lift, which is BSR minus that same seed own honest baseline. Reported at beta = 2.0 to stay comparable with the previous report.

Insights

The attack reproduces on every seed rather than being a fluke of one run: it adds +0.2415 +/- 0.0048 of backdoor lift, and letting the attackers lie about their accuracy raises that to +0.3036 +/- 0.0124, because the false claim buys them a bigger share of the merge. The full defense brings lift back to +0.0107 +/- 0.0025, a 96 percent reduction. The standard deviations are roughly fifty times smaller than the gap between the attacked and defended cases, which is the entire reason for running multiple seeds: the effect is far larger than the run-to-run noise, so the conclusion does not rest on a lucky draw. Spoofing recall is restored from 0.3641 under attack to 0.5204, close to the honest 0.5292, so the defense recovers the detector normal behaviour and not only its trigger behaviour. Clean accuracy under defense sits about 0.008 below the honest baseline, a small but consistent cost that we state rather than round away.

Honest limitation

Honest spoofing recall is only 0.5292, meaning the underlying detector misses roughly half of all spoofed samples before any attack happens, and the honest BSR is correspondingly high at 0.6367. This is a genuine weakness of the detector on this simplified dataset, and it is exactly why backdoor lift, measured against each seed own honest baseline, is the primary metric rather than raw accuracy. Improving the base detector is the highest-value work remaining.

Defense-side sensitivity (summary)

The previous report varied only the attacker side, so we swept the defense own knobs too. The result went against our expectation: a sharper penalty is worse, not safer. Backdoor lift is -0.0054 at beta = 1.0, +0.0107 at the old default of 2.0, and +0.1228 at beta = 8.0, with clean accuracy falling from 0.7106 to 0.6656 across that range. A sharp penalty punishes honest drones that merely look odd for one round, eroding the honest majority that the median-based merge depends on. This is what led directly to the fix in Section 3. The full sweep and its figure are in 1-validation/.

3. The new paper figure

Figure 1. Backdoor progression and trust attribution over federated training

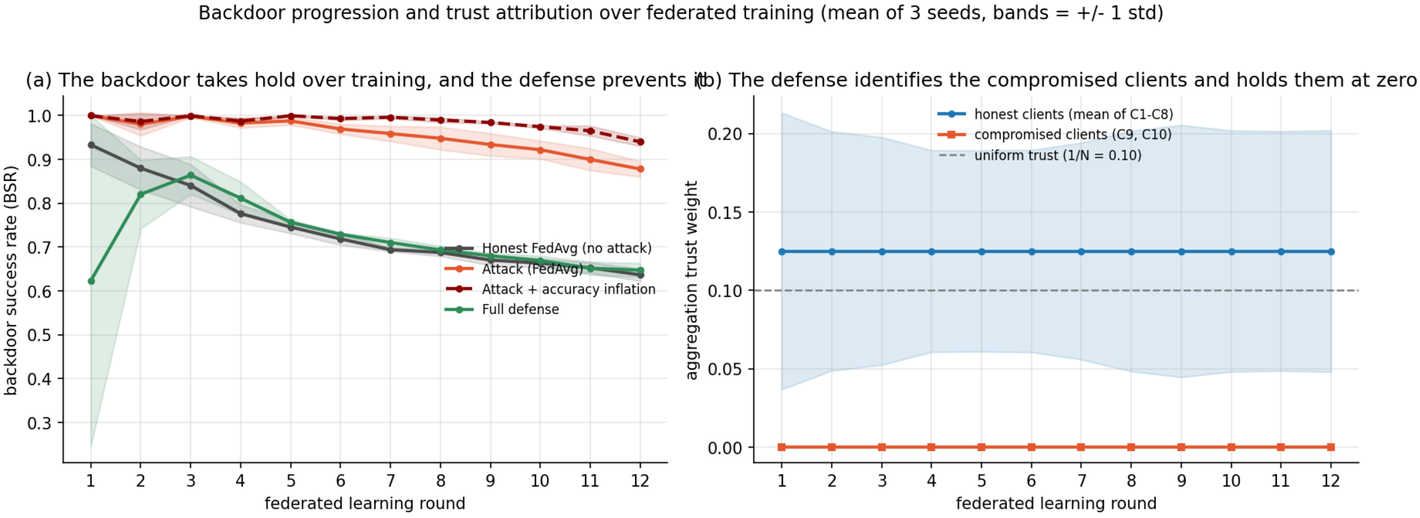

Figure 1. Panel (a): backdoor success rate on the triggered test set, measured after every federated round, for the honest baseline, the attack, the attack with accuracy inflation, and the full defense. X axis is the federated round; Y axis is backdoor success rate. Panel (b): the trust weight the server assigns to the compromised drones (C9, C10) against the honest drones (C1 to C8) under the full defense. X axis is the federated round; Y axis is trust weight; the dashed grey line marks equal trust (1/N = 0.10). All curves are means over three seeds and the shaded bands are plus or minus one standard deviation.

Insights

Panel (a) shows something a single end-of-training number cannot: the backdoor builds up gradually. The attacked curves climb away from the honest baseline round after round, and the version where the attackers also lie about their accuracy climbs faster and further, because the lie compounds their influence at every merge. The defended curve stays on the honest baseline for the whole run, so the defense is not repairing damage after the fact, it is stopping the backdoor from ever forming. Panel (b) shows the mechanism behind that flat line: within the earliest rounds the server own quiz separates the two groups and pins the compromised drones at zero trust, where they stay. Together the two panels connect cause and effect in one figure, which is the claim the paper needs: the defense works because it removes the attackers influence from the merge, and it does so almost immediately rather than gradually. The width of the honest band in panel (b) is the one visible caveat, and it is precisely the false-positive problem that Section 3 fixes.

4. Fixing the false-positive weakness (the week best result)

The checks above showed the referee wrongly suspected honest drones in 20.5 percent of client-rounds. The cause is that suspicion was penalised continuously and judged relative to the group each round, so with eight honest drones somebody is always at the bottom and a single odd round was treated like a real problem. We tested three candidate fixes, each across all three seeds.

Configuration	Clean Acc	Spoof Recall	Backdoor Lift	Honest FP rate	Attacker detection
Attack (no defense)	0.6931 +/- 0.0032	0.3633 +/- 0.0098	0.2422 +/- 0.0035	n/a	n/a
D0 Baseline (beta=2.0)	0.7029 +/- 0.0031	0.5204 +/- 0.0020	0.0107 +/- 0.0030	20.5% +/- 1.8	100.0% +/- 0.0
D1 +gentler gate (beta=1.0)	0.7106 +/- 0.0023	0.5355 +/- 0.0073	-0.0054 +/- 0.0141	6.9% +/- 1.8	100.0% +/- 0.0
D2 +dead-zone (tau=2.0)	**0.7143 +/- 0.0025**	**0.5560 +/- 0.0194**	**-0.0266 +/- 0.0173**	**0.3% +/- 0.5**	**100.0% +/- 0.0**
D3 +persistence (k=2)	0.7128 +/- 0.0023	0.5332 +/- 0.0083	-0.0005 +/- 0.0099	0.3% +/- 0.5	88.9% +/- 3.9

Table 2. Each row adds one fix to the row above. Columns: clean accuracy, spoofing recall, backdoor lift, the rate at which honest drones were wrongly flagged, and the rate at which the compromised drones were correctly flagged. Mean +/- standard deviation over three seeds. The recommended configuration D2 is highlighted. The no-defense lift here (+0.2422) differs slightly from Table 1 (+0.2415) because this ablation is a separate run of its own notebook; the gap is well inside one standard deviation.

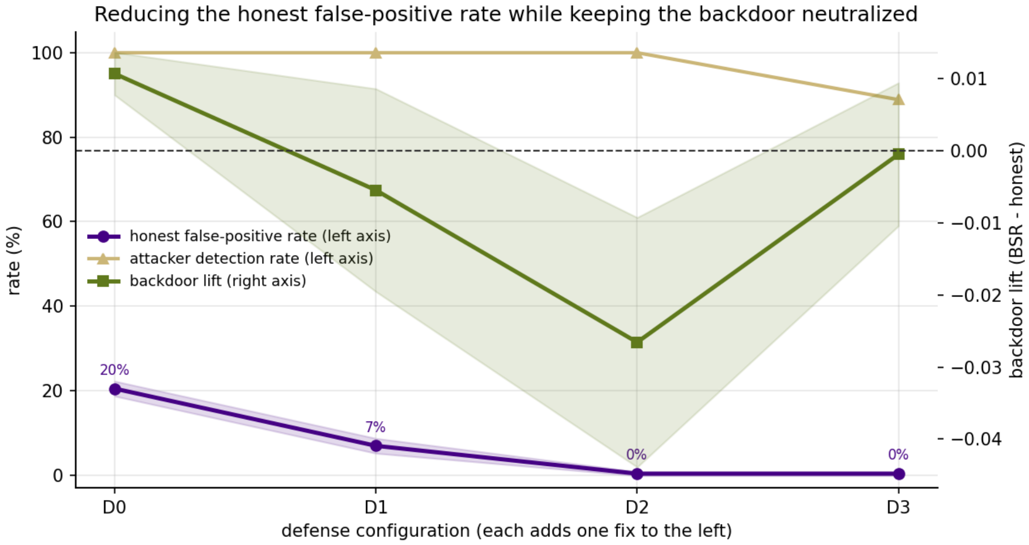

Figure 2. X axis is the defense configuration, each step adding one fix. Left Y axis is rate in percent: the honest false-positive rate (purple) and the attacker detection rate (gold). Right Y axis is backdoor lift (green), with the dashed line at zero marking the point where the attacker gains nothing.

Insights

The gentler penalty alone (D1) cuts wrongly-suspected honest drones from 20.5 to 6.9 percent, and adding the dead-zone (D2) takes it to 0.3 percent. This is not a trade: at D2 the backdoor is still neutralized, the attackers are still caught 100 percent of the time, and both utility numbers improve, with clean accuracy rising from 0.7029 to 0.7143 and spoofing recall from 0.5204 to 0.5560. D2 is therefore strictly better than the previous configuration on every metric measured, from a two-line change. The third idea, requiring a drone to look suspicious two rounds in a row before being penalised, was rejected: it did not reduce false positives any further and it dropped attacker detection to 88.9 percent, because the waiting period hands the attacker a free round. We report the measured reason rather than dropping the idea quietly.

What remains incomplete

- The base detector is weak (honest spoofing recall about 0.55), partly a ceiling of this simplified dataset. This is the highest-value next improvement.
- We stress-tested D2 against a stealth attacker that poisons less to stay hidden, and it held: below the level where the attacker becomes hard to see, its backdoor is already worthless (undefended lift +0.0051). But an attacker that explicitly optimises against the referee is untested, and is the top next experiment. Details in 3-adaptive-attacker/.
- Everything is still IID with ten drones and two attackers. Non-IID data is the natural next deeper experiment and would stress the false-positive behaviour most.
- D2 is recommended but Table 1 is still at the old beta = 2.0 for comparability. Folding D2 into the headline table is a small next task.